COMP 3500SEF · Final Report

Warehouse Logistics Management System
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Testing: Chung Ngon Man, Li Mu Qiu.

1. Scope

Goals: traceable decisions and role-separated workflows. Roles: customer, staff, driver, administrator. Local-only: one app; seed data only (RK-05, RK-10; Table B).

2. Development process

Table A. Increments

Increment and date	Trigger / activity	Outcome
Planning, 9–11 Oct 2026	Coursework kickoff	Scope, roles, API contracts (register)
17 Oct review	Stock and order review	Two defects found; fixed in row 3
Confirmation, 26–28 Oct 2026	Confirmation defects: repeat submits accepted, no stock ledger	Transaction verified (orders tests)
Integrated application, 1 Nov 2026	Role functions integrated	Customer, staff and admin workflows run end to end
Driver and privacy, 2 Nov 2026	Driver tests outstanding	Both pass (driver, privacy)
Diagram review, 10–11 Nov 2026	Diagrams diverged from the code	Aligned (review record)
Export simplification, 11 Nov 2026	Admin export took four interactions (issue #15)	Selected-date export in three interactions (fill both dates, export)
CI health tests, 11–12 Nov 2026	No test covered CI-run outcomes	Two regression tests; suite reached 38 (unit.log)

3. Requirements

- FR-01 create orders with line items Test. - FR-02 enforce lifecycle, stock release Test (cancel). - FR-03 number orders and shipments; track the order holder Test. - FR-04 inventory, ledger, low-stock alerts Test. - FR-05 record, read notifications Test. - FR-06 admin aggregates, CSV Test. - FR-07 authenticate, access Test. - FR-08 assign, deliver Test.

- NFR-01 performance: API p95 ≤2 s, first contentful paint p95 ≤3 s on the local host (measurement). - NFR-02 CI health — every commit ends in green unit, integration, browser runs (runs, observations). - NFR-03 least privilege: role gates; delivery address only to the assigned driver (privacy tests). - NFR-04 interaction budget: every core operation (the eleven in §8) takes at most three in-app interactions (one = one fill, selection or click; navigation excluded) (walkthrough). - NFR-05 process quality — every increment closes on a recorded review or test run.

Table B. Risk coverage (RK = register IDs)

Risk	Addressed by	Evidence
①RK-01/03/06/08 schedule, test week, coursework	weekly re-planning, scope priorities	risk plan
②RK-02 contract rework	lifecycle locked as test constants	constants tests
③RK-04/07 setup, stack	setup guide; CI per commit	README ci.yml
④RK-05 demo data	deterministic seed, regression-tested	seed test
⑤RK-09/10 hosting; personal data	team decision, 9 Oct — seed data only, no real personal data; NFR-03 governs in-app access	risk plan

4. System models

Layers: pages, routes, handlers, Prisma/SQLite. **Order items** reference one product; orders belong to one user; an order has at most one shipment; products accumulate movements (InventoryLog); notifications tie to at most one order.

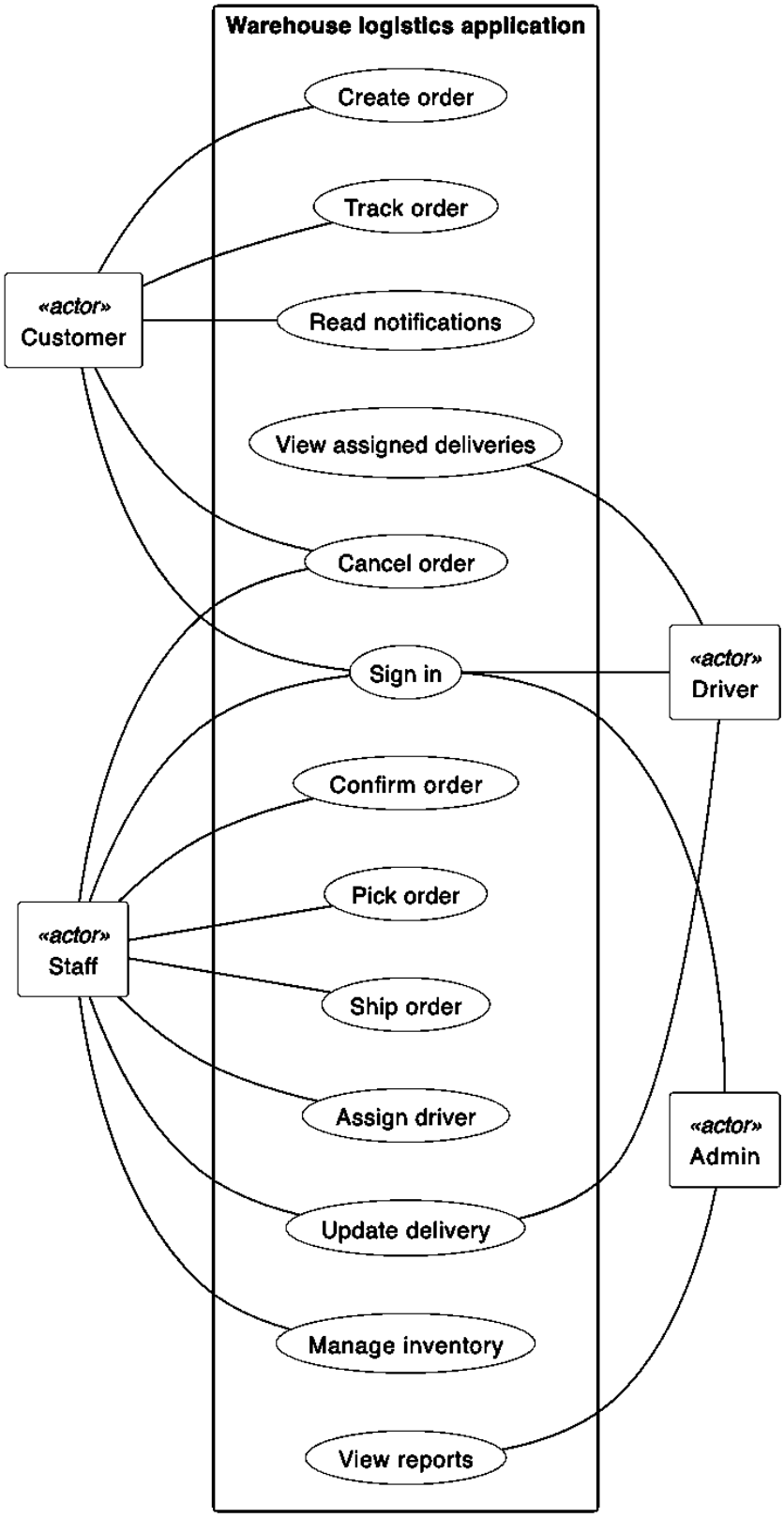


Figure 1. Thirteen use cases under four roles.

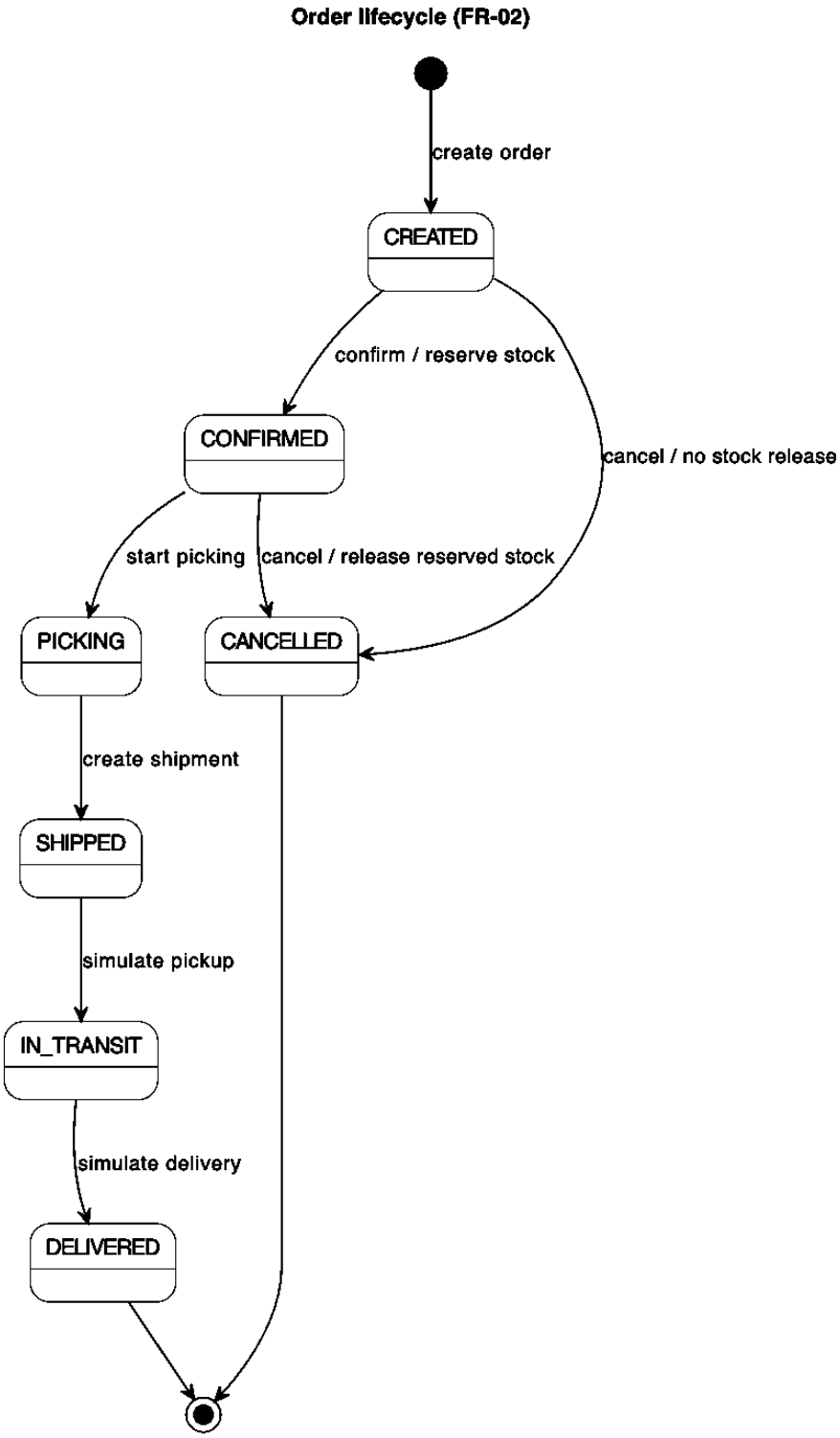


Figure 2. Order transitions: which reserve stock, which release it, which leave it unchanged.

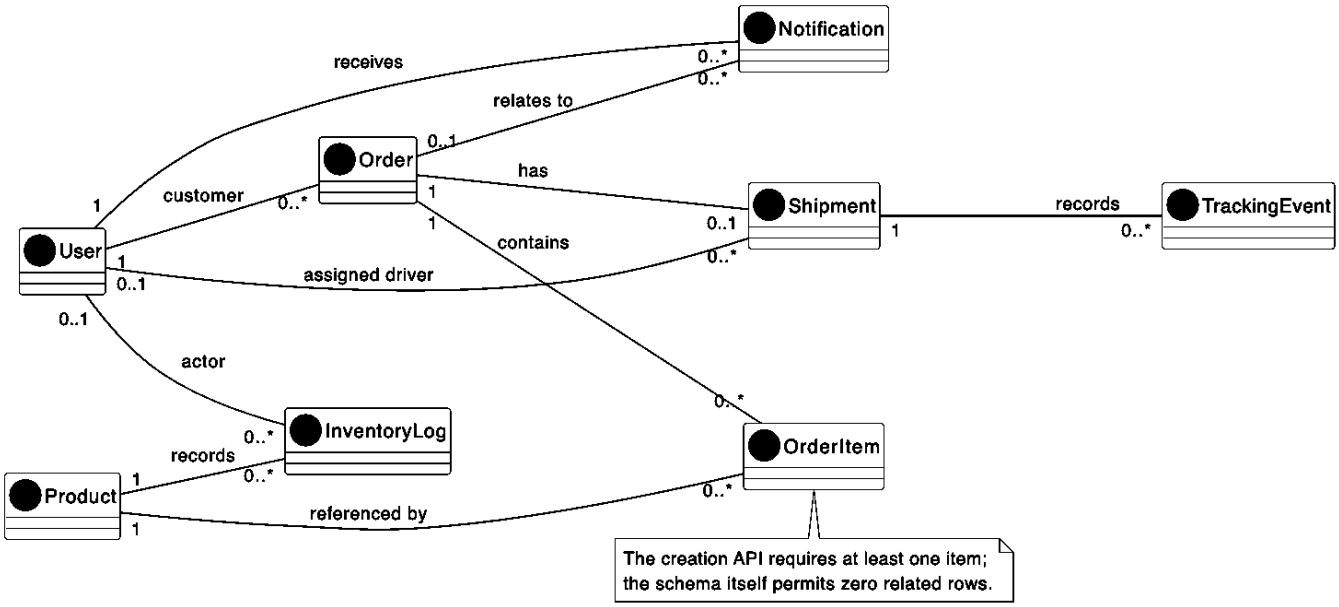


Figure 3. Entities and multiplicities.

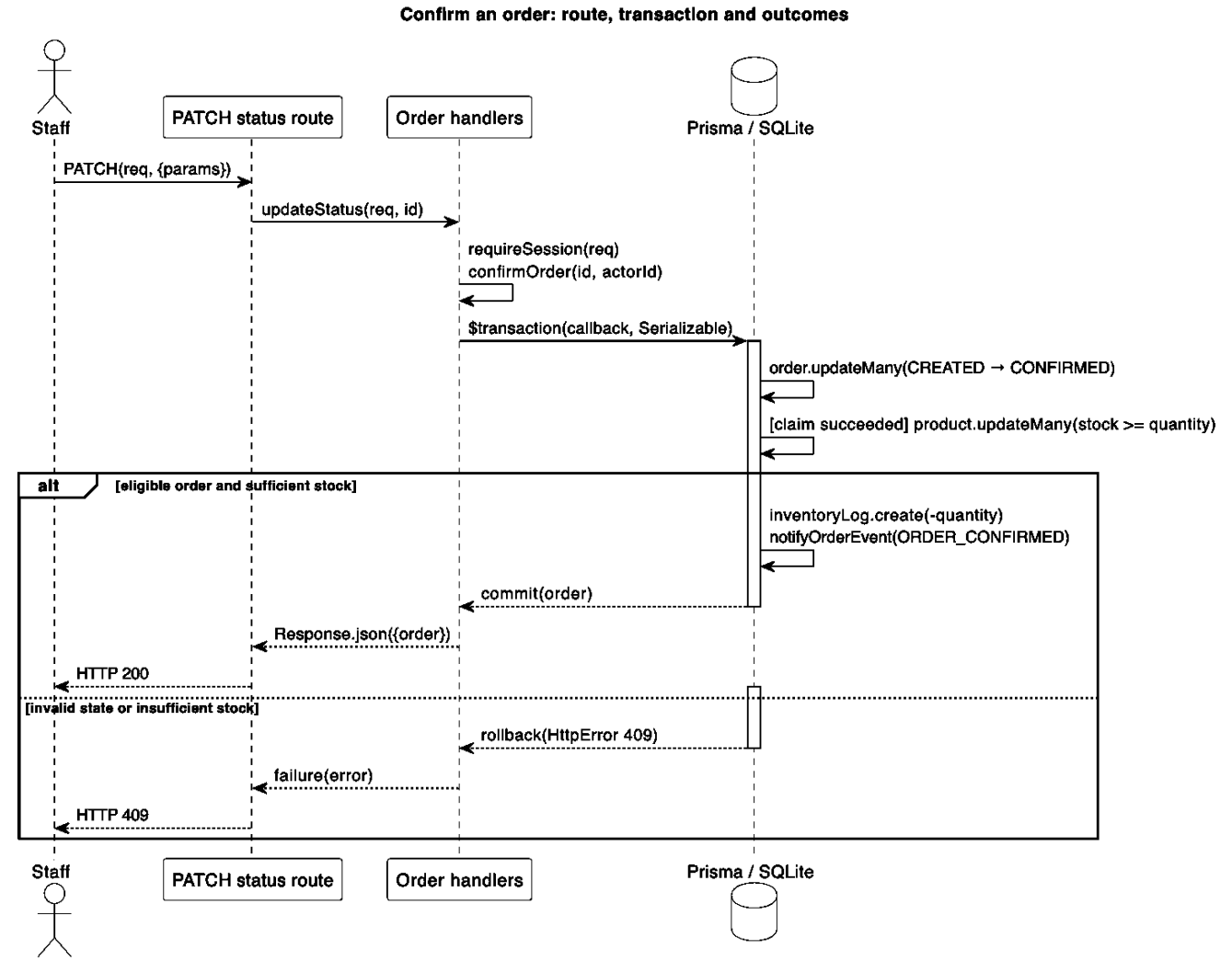


Figure 4. Confirmation: authenticated session, conditional writes, rollback.

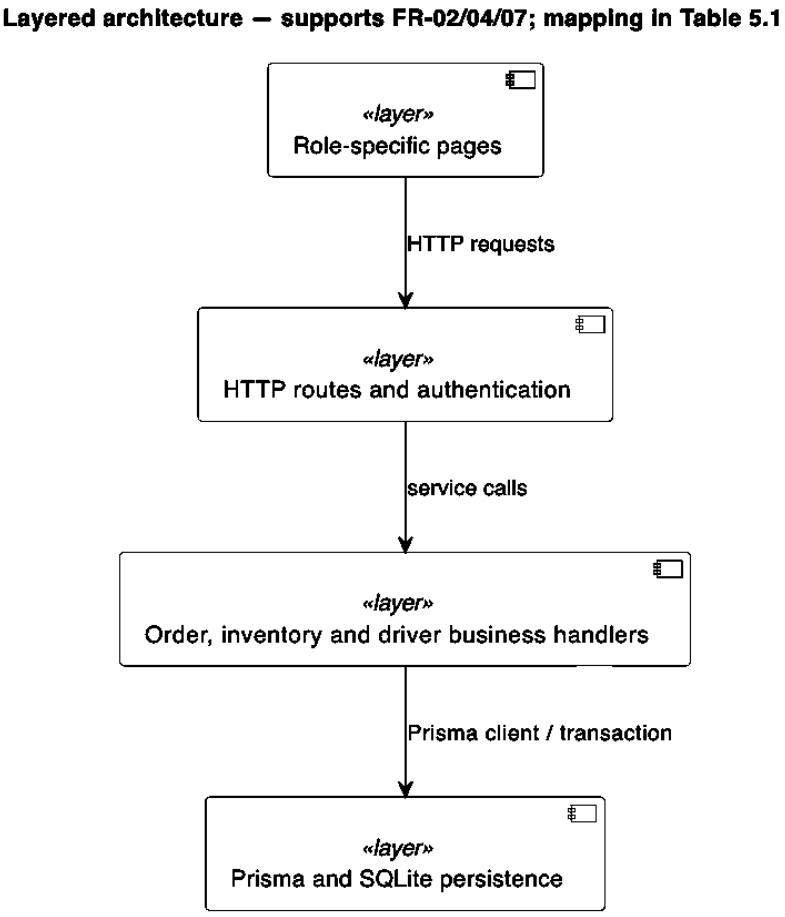


Figure 5. The four layers.

5. Implementation

Confirmation reserves stock; lifecycle CREATED→CONFIRMED→PICKING→SHIPPED→IN_TRANSIT→DELIVERED; cancel from CREATED/CONFIRMED.

Cancellation permissions:

Role	Transition and effect
Customer	CREATED: cancel, no restock
Customer	CONFIRMED: cancel, restock
Staff	CREATED: rejected
Staff	CONFIRMED: cancel, restock
Customer, staff	PICKING onwards: rejected

A CREATED order advances only by the customer's cancel or the staff confirmation.

Atomic confirmation (HTTP 409 on repeats; rollback on short stock); driverId set at SHIPPED, immutable after; FR-03/04 verified (order-api, driver-api, inventory-api).

6. Verification

Verification checks the implementation against §3; validation exercises the delivered workflows (§8). CI records 38 unit, 140 integration, 2 integrity checks: the demo-data reset requires RESET_DEMO_DATA=YES (evidence protection), and the packaged submission's SHA-256 manifest must match the source (integrity checks). Logs: integration "tests 140", "pass 140"; unit "Tests 38 passed (38)".

Loopback measurement: p95 3.2–13.6 ms, first contentful paint p95 64–96 ms (30 samples per endpoint) vs 2/3 s — loopback, not representative of public networks.

Reviews (17 Oct, 10–11 Nov, 11 Nov) statically verified requirements and models; the export fix was then validated through the interface (§8).

7. Evolution

Reviews: 17 Oct found two defects (repeat submits, missing ledger) fixed by the confirmation transaction (§5); 10–11 Nov aligned diagrams and rewrote NFR-01 (loopback measurement) and NFR-02 (daily window → CI-run checks); the 11 Nov review added the direct path (issue #15). Feedback that the CI record was untested added two regression tests (row 8; suite 36→38).

8. Validation

Walkthrough covers eleven core operations: create order; open details; confirm; start picking; ship; assign driver; record pickup; record delivery; filter report; export current; export selected dates — at most three interactions each; the 26 checks reach all thirteen of Figure 1's use cases (coverage map). "Browser acceptance passed: 26 checks" (browser log, the walkthrough run): 20 checks (five operations once, three twice, three thrice) plus 6 cross-cutting checks — validation, session, interface: two each. During report preparation, the two testers (Chung Ngon Man, Li Mu Qiu) repeated the six core tasks — 64 of 64 attempts unaided, median 2.1 min per session (user-validation).

9. Feedback and reconciliation

Sources: whole-team reviews; test runs and the scripted walkthrough executed by Chung Ngon Man and Li Mu Qiu (Testing).

Table C. Feedback

Feedback (source, date)	Disposition and evidence
Stock effects listed under FR-04 (17 Oct, traceability)	Moved to FR-02 (§3)
Thin triggers (10–11 Nov, register)	Rewritten
Filter-then-export took four interactions (11 Nov, issue #15)	Rejected — combined steps already totalled four; the direct path (fill both dates, export) meets NFR-04; tracked in issue #15
No CI-health test (11 Nov, unit.log)	Two regression tests (row 8); suite 38
NFR-01/02 assumed hosted measurement, daily window (10–11 Nov, revision log)	NFR-01 → loopback; NFR-02 → CI-run checks (§6)

10. Evidence

38 unit, 140 integration, 2 integrity (12 Nov), 26 browser checks (11 Nov); after it, only row 8's regression tests landed (history) — BASELINE. Traceability.